
MATH 525

Probability

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Recommended Reading: Probability and Random Processes (Grimmett/Stirzker)

Mathematical study of randomness (random $\leftrightarrow$ deterministic)

How do we model Randomness?

Coin: $P(H) = P(T) = 0.5$

Biased Coin: $P(H) = 0.7, P(T) = 0.3$

Dice: $P(1) = P(2) = \dots = P(6) = \frac{1}{6}$

$P(1 \text{ or } 3) = P(\{1, 3\}) = P(1) + P(3) = \frac{1}{3}$

Power Law: $P(n) = \frac{6}{\pi^2} \frac{1}{n^2}$

$P(n \text{ is even}) = P(\{2, 4, 6, \dots\}) = \frac{1}{2}$

$P(4 \text{ divides } n) = P(\{4, 8, \dots\})$

Sample space and probability measure

Definition 0.1.1 (Sample Space). Let Ω be a set. Let F be a collection of subsets of Ω , which is a σ -algebra

1. $\emptyset \in F$
2. If $A \in F$, then $A^c \in F$
3. If $A_1, A_2, \dots \in F$ then $\bigcup_n A_n \in F$

Definition 0.1.2 (Probability Measure). A Probability measure on (Ω, F) is a function

$\mathbb{P} : F \mapsto [0, 1]$ s.t.

- $\mathbb{P}(\emptyset) = 0, \mathbb{P}(\Omega) = 1$
- If $A_1, A_2, \dots \in F$ is disjoint collection then $\mathbb{P}(\bigcup_n A_n) = \sum_n \mathbb{P}(A_n)$

Exercise 0.1.3. • $A, B \in$